equivalent embedding (ignoring the labels) regardless of which edge we remove from K_5.

In conclusion, the largest sub-graphs of K_5 are planar, and hence any other graph with 5 or less vertices and less than 9 edges is planar.

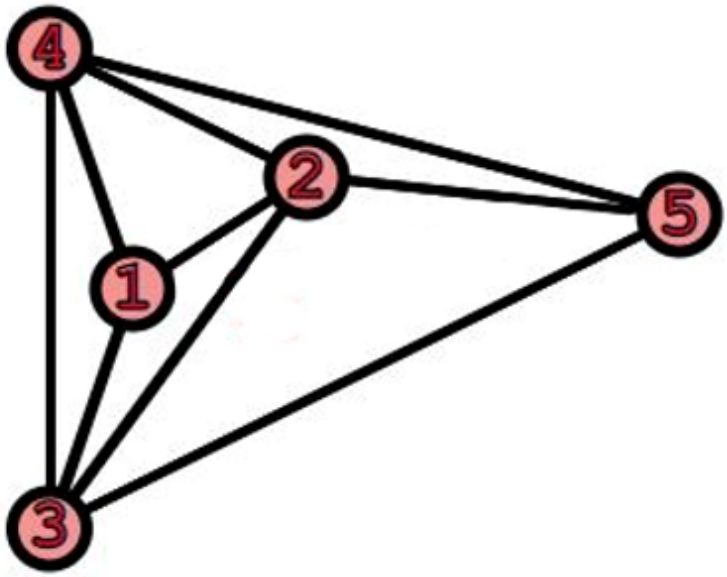


Figure 15.9 Any graph obtained by removing an edge from K_5 can be embedded in the plane without any intersection.

The same thing can be shown for $K_{3,3}$: investigating its sub-graphs can be a good exercise to grasp a better understanding of bipartite graphs and embeddings.

15.2.2 Planarity Testing

Checking if a graph is planar is trickier than you might think. Even checking if an embedding is planar isn't that easy: that's one of the tasks our brain performs easily, but it's not